

Are we Prepared Enough for AI?

During the past several months a lot has been happening in the world of artificial intelligence. AI has shifted from being new and excited about all the new capabilities, to being very serious about the risks, governance and whether organizations are truly prepared for the technology they're adopting in the workplace. Through some research I was able to find three articles; an IEEE article on the "AI readiness gap," a report from Oak Ridge National Laboratory on the Photon vulnerability discovery framework, and an ACM Europe policy brief on systemic risks from agentic AI, all point to the same underlying issue: AI is moving faster than the structures meant to keep it safe (IEEE InSight, 2024; Oak Ridge National Laboratory, 2024; ACM Europe Technology Policy Committee, 2024). Together, they paint a picture of a technology ecosystem where capability is accelerating, but understanding, oversight, and policy are struggling to keep up.

The biggest issue found in each article is that AI systems are becoming more autonomous and more deeply embedded in critical processes, yet most organizations and governments still lack the frameworks, training, and technical tools to manage all of the risks that come with it. The IEEE article highlights a very practical problem: companies are adopting AI tools without giving employees the knowledge or context to use them responsibly (IEEE InSight, 2024). Workers are being told to "use AI to be more productive," but they're not being taught how the systems work, what their limitations are, or how to recognize mistakes AI can make, not totally relying on the AI. The gap between adoption and understanding creates a grave danger to the business, allowing for more mistakes, misuse, and over-reliance in the work place.

This same issue came up in my professional writing class just this past Thursday. We had a full debate about how AI is going to change education and writing, and my stance lined up almost exactly with what these articles are warning about. I argued that once you're in college, every class should include a module, focused around 3/4ths into the semester, that teaches students how to properly use AI in that specific course. By this point, you've already learned the core material, so you're not using AI to replace learning; you're using it to enhance what you already understand. Teaching proper usage reduces the temptation to cheat and instead frames AI as a tool, not a shortcut. And because each major uses AI differently, learning the right way to use it in each class prepares you for how it will show up in your career after graduation. That classroom debate made it even clearer to me that the "readiness gap" isn't just a workplace issue; it starts in education.

On another note, the ORNL article shows the opposite side of things, the high technical risks that most companies/or organizations aren't even aware of. Their Photon framework is designed to uncover vulnerabilities in AI models at scale, including weaknesses that could be exploited or that could cause models to behave unpredictably (Oak Ridge National Laboratory, 2024). Why there is even a need to build stuff like this should say a lot: AI systems attack surfaces and failure modes that are not obvious, and traditional cybersecurity ways aren't enough

and working. This just reinforces the idea that AI risk isn't just about bias or misinformation; it's also about robustness, security and the possibility of systematic failures.

The ACM policy brief article goes in even deeper, focusing on “agentic AI”; systems that can take actions, make decisions and operate with a degree of autonomy. The brief argues that these systems introduce systemic risks that current regulations, including the EU AI Act, aren't fully prepared to address (ACM Europe Technology Policy Committee, 2024). The biggest concern isn't just that an AI system might make a mistake; it's that interconnected, autonomous systems could create cascading effects across sectors. When AI agents can initiate actions, interact with other systems, and adapt their behavior, the potential for unintended consequences grows dramatically. This is where policy gaps become dangerous.

From a policy perspective, these three sources point toward several implications. First, voluntary guidelines and high-level principles aren't enough. Organizations need enforceable standards for AI training, documentation, testing, and monitoring. Second, governments will need to expand their regulatory frameworks to address not just model outputs, but model behavior, autonomy, and interactions with other systems. Third, technical tools like Photon will likely become part of regulatory expectations; similar to how penetration testing became standard in cybersecurity (Oak Ridge National Laboratory, 2024). And finally, the policy landscape will need to evolve continuously, because AI capabilities are evolving continuously.

The stakeholders involved in this issue are broad: government agencies, private companies, researchers, workers, and the general public. Regulators are trying to keep up with a moving target. Companies want to innovate quickly but often lack internal governance structures. Researchers are uncovering new risks faster than policymakers can respond. Workers are being asked to use tools they don't fully understand (IEEE InSight, 2024). And society is left dealing with the consequences when systems fail or behave unpredictably.

My own assessment is that the real problem isn't AI itself; it's the mismatch between capability and readiness. Organizations are excited about what AI can do, but they're not building the guardrails, training, or accountability structures needed to use it safely. At the same time, policymakers are trying to regulate a technology that doesn't sit still long enough for traditional legislative processes. That tension creates a risky environment where AI systems are deployed widely before anyone fully understands their implications. And based on the debate in my writing class, even students feel that gap, we're using AI more and more, but we're not always being taught how to use it responsibly or effectively.

For information professionals, this shift means their roles are becoming even more critical. They'll need to understand not just data quality and system design, but also AI behavior, model limitations, and risk mitigation strategies. They may be responsible for evaluating AI tools, documenting their use, or ensuring compliance with emerging standards. For society, the implications are even bigger. AI systems influence hiring, healthcare, security, and access to

information. If these systems are deployed without proper oversight, the consequences could be widespread and long-lasting.

In the end, these three sources collectively highlight a simple but important truth: AI governance can't be an afterthought. As AI becomes more capable and more autonomous, the need for thoughtful, informed, and proactive policy becomes impossible to ignore.

References

ACM Europe Technology Policy Committee. (2024). *Systemic risks associated with agentic AI* [Policy brief]. https://www.acm.org/binaries/content/assets/public-policy/europe-tpc/systemic_risks_agentic_ai_policy-brief_final.pdf

IEEE InSight. (2024). *The AI readiness gap: Why adoption without understanding fails.* <https://insight.ieeeusa.org/articles/the-ai-readiness-gap-why-adoption-without-understanding-fails/>

Oak Ridge National Laboratory. (2024). *Photon framework scales AI vulnerability discovery.* <https://www.ornl.gov/news/photon-framework-scales-ai-vulnerability-discovery>